

Vaishnavi Sadanand Tandel

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Computer Science Engineer | Full Stack Developer | Data Analyst

PROFESSIONAL SUMMARY

Computer Science Engineering graduate (CGPA: 8.74) with hands-on experience in full-stack development, REST API development, databases, and web technologies. Skilled in building scalable applications using JavaScript, Python, MERN stack, and SQL with a strong foundation in problem-solving and software development practices.

EDUCATION

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| • Sahyadri College of Engineering and Management, Mangalore (VTU)
<i>Bachelor of Engineering in Computer Science and Engineering</i> | <i>December 2022 – May 2026</i>
CGPA: 8.74 / 10.0 |
| • Government PU College Aversa, Uttara Kannada
<i>Pre-University (Class XII) – Science (PCMB), Karnataka State Board</i> | <i>May 2020 – April 2022</i>
Percentage: 85.5% |

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript (ES6+), SQL, HTML5, CSS3
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Flask
- **Databases:** MongoDB, MySQL
- **Tools & Platforms:** Git, GitHub, Postman, VS Code, Claude Code
- **Core Competencies:** REST APIs, JSON, Database Modeling, Problem Solving

WORK EXPERIENCE

- **GAwC Renewables LLP**
Wind Data Analysis Intern Feb 2026 – May 2026
 - Analyzed and processed ERA5 wind datasets using Python and pandas for trend and performance evaluation.
 - Converted climate data into structured CSV reports and created visualizations for analysis.
 - Automated data processing workflows to improve efficiency and reduce manual effort.
 - Developed an ERA5 Wind Data Processing Tool using Flask and Python for data retrieval and report generation.

KEY PROJECTS

- **Multimodal Deep Learning Framework for Early COPD Prediction** | [GitHub](#)
Technologies: Python, TensorFlow/Keras, CNN-LSTM, MLP, MFCC, Librosa, Flask, NumPy, Pandas
 - Contributed to an AI-based diagnostic system integrating spirometry and lung audio data, achieving 98
 - Built and trained CNN-LSTM and MLP models for respiratory audio and clinical data analysis.
 - Deployed the solution as a Flask web application for real-time COPD risk prediction.
 - Preprocessed and optimized datasets to improve model performance and stability.
- **Adventure Trekking Platform with Booking & Admin Management System** | [GitHub](#) | [Live Demo](#)
Technologies: MongoDB, Express.js, React.js, Node.js, JavaScript, Tailwind CSS, Nodemailer, REST APIs, JWT
 - Developed a full-stack MERN application with Admin and User roles for trek and booking management.
 - Implemented booking workflows with real-time seat availability and automated email notifications.
 - Designed 15+ REST APIs with JWT authentication for secure client-server communication.
 - Optimized MongoDB queries for efficient data handling and performance.
- **ERA5 Wind Data Processing Tool** | [GitHub](#) | [Live Demo](#)
Technologies: Python, Flask, SQLite, HTML, CSS, JavaScript, xarray, pandas
 - Developed a full-stack web tool to retrieve and process ERA5 wind data from the Copernicus Climate Data Store (CDS).
 - Implemented JWT authentication, API integration, and automated NetCDF/CSV report generation.
 - Built data processing features for wind speed, wind direction, and power density analysis.

CERTIFICATIONS

- **Database & SQL Course Completion** | [Certificate](#) Feb 2025
 - Learned database architecture, normalization, advanced SQL, and query optimization techniques.
- **Data Analytics Job Simulation (Deloitte Australia)** | [Certificate](#) Oct 2025
 - Applied data analysis skills and created dashboards using Tableau and Excel for business insights.